WHY INDOOR FARMS?

Labor Intensive with too many Fragmented Systems

Manual Systems such as
- Carbon Dioxide Supply Unit
- Lighting
- Form factors & Growing Systems
- Environmental Control Unit
- Air Conditioning
- Nutrient Supply & Control Unit

Savings Potential through increased Human Productivity

Utilities	$
Rent	$ $
Human Capital	$ $ $

Undesired Interactions between Building Conditions and Crops

External building conditions like temperature affects crops.

- Inconsistent Growing Environment
- Habitat prone to Mold and Pests
- Affect level of Photosynthesis

BUILDING MANAGEMENT SOLUTION FOR INDOOR FARMS

Ambient Systems coordinate complex parameters, such as lighting, moisture, and CO2, requiring constant monitoring and adjustments to ensure *high-quality* output for Indoor Farms.

- **Sense**
Seamless integration of sensors, smart meters, smart devices, and assets.

- **Optimize**
Building information modelling (BIM) with operational twin combining digital model of building design and live automation data.

- **Control**
Shift from diagnostic and descriptive manual controls to predictive and automated ones.

Approved by industry leaders in building technology.

HOW IT WORKS

1. Connected

Integrated with essential equipment and control systems

Seamless Building Integration
Connect the indoor farming system to the existing building management system – or have Ambient implement them based on your requirements. Enjoy full connectivity across the farm and perform real-time environmental adjustments based on crop.

2. Centralized

Important data in one easy-to-access place

Monitoring In Real-Time
Access real-time data through online dashboards and securely archive all critical operational information. Start measuring most important crop parameters like temperature, relative humidity, CO2, and light intensity.

3. Automated

Machine Learning-enabled optimization

Grow With Data Insights
Leverage latest machine learning models to improve the harvest and reduce operational costs. Apply data-driven solutions to understand key factors that affect plant growth and quality, enabling better crop health and more accurate harvest timing predictions.

DATA DRIVEN AUTOMATION

Climate Data
- Air Temperature
- Relative Humidity
- Carbon Dioxide (CO2)
- Volatile Organic Compounds (VOCs)

Crop Data
- Soil Temperature
- Soil Moisture
- pH level
- Electrical conductivity (ECw)
- Crop specific parameters

Equipment Settings
- Adjust temperature
- Adjust ventilation
- Adjust lights
- Install fogging machines
- Install CO2 regulator
- Increase light intensity
- Increase natural light

+24% Energy Savings, Man-Hour Savings, Reduction in Crop Loss

Break-even in 6 to 12 Months

ABOUT AMBIENT SYSTEMS

Ambient Systems is a deep-tech spin-off from the National University of Singapore (NUS) that is developing an integrated hardware-software solution for climate management of vertical farms.

The product leverages intellectual property on securely combining various sources of data and machine learning to enhance building performance.

Approved by Industry Leaders in Building Technology, Ambient Systems work with high-value crop farms.

PARTNERS

National University of Singapore	Singrow

TEAM

Product & Engineering

Wa Low
Design & Marketing

Get in touch with us

If you have any questions or if you are...

Indoor Farms
interested in building automation solutions.

Advisors
with expertise in Agri-tech, [redacted], Building Technology.

Technology Partners
specializing in sensor hardware and system integration.

First Name

Last Name

Company Name

Phone Number

Email

Message

Submit

info@ambientsystems.ai

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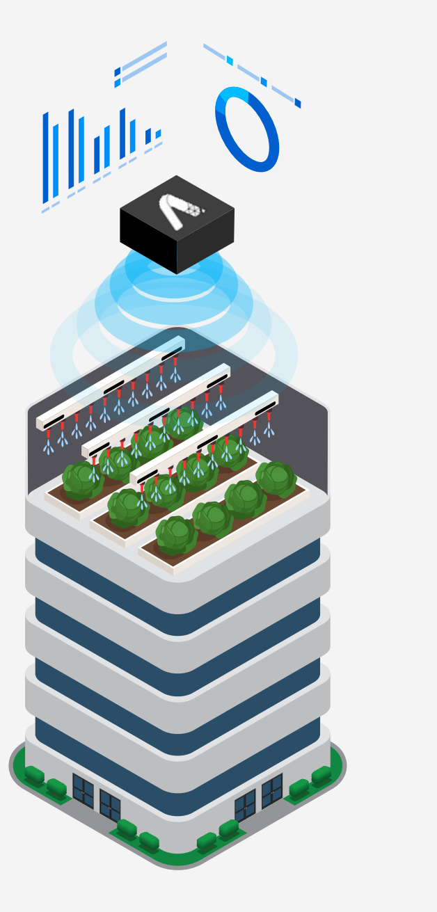

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